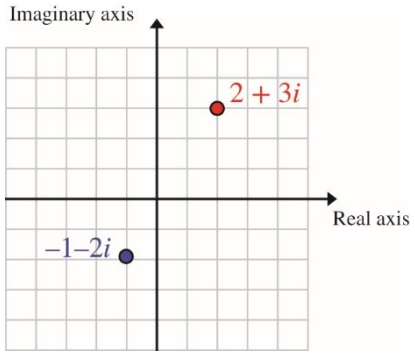
## Plot Complex Numbers in the Complex Plane

A complex number $a + bi$ has two parts, the real part $a$ and the imaginary part $b$. Two axes are required to represent $a + bi$ graphically.

Real axis: the horizontal axis represents the real part of a complex number
Imaginary axis: the vertical axis represents the imaginary part
The real and imaginary axes together form the **complex plane**.




A complex number $z = a + bi$ is written as an ordered pair $(a, b)$ and can then be graphed as such in the complex plane.

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Graph the complex numbers in the complex plane.

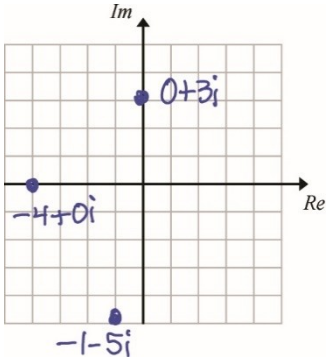
$-1 - 5i$ $(-1, -5)$

1.

$3i$ $(0, 3)$

2.

3. $-4$ $(-4, 0)$

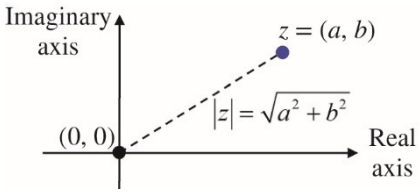



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The absolute value of a complex number $z = a + bi$ is the distance between $(a, b)$ and $(0, 0)$ in the complex plane.

The absolute value of a complex number $|z|$ is more commonly called the modulus (or magnitude) of the number and $|z| = \sqrt{a^2 + b^2}$.

Find the modulus




$z = -7 - 12i$

$|z| = \sqrt{(-7)^2 + (-12)^2}$

4. $= \sqrt{193}$

$z = 4 - 6i$

5. $|z| = \sqrt{4^2 + (-6)^2} = 2\sqrt{13}$

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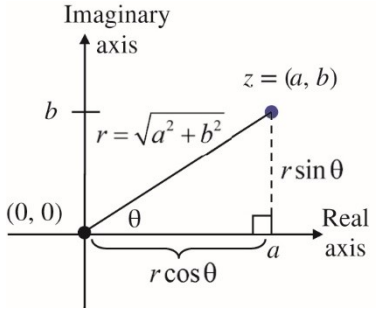
## Write the Polar Form of a Complex Number

The complex number $z = a + bi$ is said to be in rectangular (or Cartesian) form because $a$ and $b$ are the coordinates of the point $(a, b)$ in the rectangular coordinate system formed by the complex plane.

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The polar form (or trigonometric form) of a complex number $z = a + bi$ is

$$z = r(\cos\theta + i\sin\theta)$$




where $a = r\cos\theta,\ b = r\sin\theta,\ r = \sqrt{a^2 + b^2},\quad \tan\theta = \dfrac{b}{a}\ (a \neq 0).$ and

$$z = a + bi\ = r\cos\theta + i \cdot r\sin\theta = r(\cos\theta + i\sin\theta)$$

The value of $r$ is the modulus of $z$ and we say that $\theta$ is an *argument* of $z$.

There are infinitely many choices for $\theta$, but normally $\theta$ is taken on the interval $0 \le \theta < 2\pi$ or $0° \le \theta < 360°$.

$z = r(\cos\theta + i\sin\theta)$ is sometimes shortened to $z = r\text{cis}\,\theta$ where "cis" stands for "cosine *i* sine."

Example: $z = 2\left(\cos\dfrac{\pi}{3} + i\sin\dfrac{\pi}{3}\right)$ can be written as $z = 2\text{cis}\dfrac{\pi}{3}$

Additionally, for a real number $\theta$, Euler's formula states that

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

Example: $z = 2\left(\cos\dfrac{\pi}{3} + i\sin\dfrac{\pi}{3}\right)$ can be written as $z = 2e^{\frac{\pi}{3}i}$

---

Write the complex number in polar form.

6. $z = -\dfrac{\sqrt{2}}{2} - \dfrac{\sqrt{2}}{2}i$ (in Q III)

$$r = \sqrt{\left(-\frac{\sqrt{2}}{2}\right)^2 + \left(-\frac{\sqrt{2}}{2}\right)^2} = 1$$

$$\tan\theta = \frac{-\frac{\sqrt{2}}{2}}{-\frac{\sqrt{2}}{2}} = 1,$$

$$\theta = \frac{5\pi}{4}$$

$$z = \cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}$$

7. $z = 12 + 5i$ (in QI)

$$r = \sqrt{12^2 + 5^2} = 13$$

$$\tan\theta = \frac{5}{12},$$

$$\theta = \tan^{-1}\frac{5}{12}\ (\approx 0.39 \text{ or } 22.6°)$$

$$z = 13\left(\cos\left(\tan^{-1}\frac{5}{12}\right) + i\sin\left(\tan^{-1}\frac{5}{12}\right)\right)$$

$$\approx 13(\cos 0.39 + i\sin 0.39)$$

$$\text{or} \approx 13(\cos 22.6° + i\sin 22.6°)$$

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8. Write $z = 6\left(\cos 135° + i \sin 135°\right)$ in rectangular form, $a + bi.$

$$z = 6\left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\right) = -3\sqrt{2} + 3\sqrt{2}i$$

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Compute Products and Quotients of Complex Numbers in Polar Form

## Products and Quotients of Complex Numbers in Polar Form

If $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$ then,

$z_1 z_2 = r_1 r_2\left[\cos(\theta_1+\theta_2)+i\sin(\theta_1+\theta_2)\right]$ and $\dfrac{z_1}{z_2} = \dfrac{r_1}{r_2}\left[\cos(\theta_1-\theta_2)+i\sin(\theta_1-\theta_2)\right]$ $(z_2 \neq 0)$.

---

9. Given $z_1 = 20(\cos 30° + i\sin 30°)$ and $z_2 = 3(\cos 210° + i\sin 210°)$ find $z_1 z_2$ and write the product in polar form.

$$z_1 \cdot z_2 = 20\cdot 3(\cos(30°+210°)+i\sin(30°+210°))$$
$$= 60(\cos 240° + i\sin 240°) \quad \text{or} \quad 60\ \text{cis}\ 240°$$

10. Given $z_1 = 8\left(\cos\dfrac{7\pi}{12} + i\sin\dfrac{7\pi}{12}\right)$ and $z_2 = 2\left(\cos\dfrac{\pi}{4} + i\sin\dfrac{\pi}{4}\right)$, find $\dfrac{z_1}{z_2}$ and write the quotient in polar form.

$$\frac{z_1}{z_2} = \frac{8}{2}\left(\cos\left(\frac{7\pi}{12}-\frac{\pi}{4}\right)+i\sin\left(\frac{7\pi}{12}-\frac{\pi}{4}\right)\right)$$
$$= 4\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right) \quad \text{or} \quad 4\ \text{cis}\frac{\pi}{3}$$

11. Given $z_1 = \dfrac{1}{2} + \dfrac{\sqrt{3}}{2}i$ and $z_2 = \dfrac{\sqrt{3}}{2} + \dfrac{1}{2}i$ find $z_1 z_2$ and write the product in polar form.

$z_1$ in Q I

$$r = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = 1$$

$$\tan\theta = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3},\ \theta = \frac{\pi}{3}$$

$$z_1 = \text{cis}\frac{\pi}{3}$$

$z_2$ in Q I

$$r = \sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = 1$$

$$\tan\theta = \frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}},\ \theta = \frac{\pi}{6}$$

$$z_2 = \text{cis}\frac{\pi}{6}$$

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$$z_1 \cdot z_2 = \text{cis}\left(\frac{\pi}{3} + \frac{\pi}{6}\right) = \text{cis}\frac{\pi}{2} = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2}$$

i.e. $z_1 z_2 = i$

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## Compute Powers of Complex Numbers (De Moivre's Theorem)

**De Moivre's Theorem** If $z = r(\cos\theta + i\sin\theta)$ and $n$ is a positive integer, then

$$z^n = [r(\cos\theta + i\sin\theta)]^n = r^n(\cos n\theta + i\sin n\theta).$$

---

12. Compute $\left[2\left(\cos\frac{3\pi}{16} + i\sin\frac{3\pi}{16}\right)\right]^4$ and write the result in rectangular form $a + bi$.

$$\begin{aligned}\left[2\left(\cos\frac{3\pi}{16} + i\sin\frac{3\pi}{16}\right)\right]^4 &= 2^4\left(\cos\left(4\cdot\frac{3\pi}{16}\right) + i\sin\left(4\cdot\frac{3\pi}{16}\right)\right)\\ &= 16\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)\\ &= 16\left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right)\\ &= -8\sqrt{2} + 8\sqrt{2}i\end{aligned}$$

13. Compute $\left(2\sqrt{3} + 2i\right)^6$ and write the result in rectangular form $a + bi$.

$$r = \sqrt{(2\sqrt{3})^2 + 2^2} = 4,\ \tan\theta = \frac{2}{2\sqrt{3}} = \frac{1}{\sqrt{3}} \Rightarrow \theta = \frac{\pi}{6} \quad (\text{QI})$$

$$\begin{aligned}(2\sqrt{3} + 2i)^6 &= \left[4\,\text{cis}\frac{\pi}{6}\right]^6\\ &= 4^6\,\text{cis}\left(6\cdot\frac{\pi}{6}\right)\\ &= 4096\,\text{cis}\,\pi\\ &= 4096(\cos\pi + i\sin\pi)\\ &= 4096(-1 + 0)\\ &= -4096\end{aligned}$$

---

## Find the *n*th Roots of a Complex Number

**nth Roots of a Complex Number**

If $z = r(\cos\theta + i\sin\theta)$, then for an integer $n \quad 2$, $z$ has $n$ distinct $n$th roots of the form:

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$$w_k = \sqrt[n]{r}\left[\cos\left(\frac{\theta + 2\pi k}{n}\right) + i\sin\left(\frac{\theta + 2\pi k}{n}\right)\right] \qquad w_k = \sqrt[n]{r}\left[\cos\left(\frac{\theta + 360°k}{n}\right) + i\sin\left(\frac{\theta + 360°k}{n}\right)\right]$$

or

for $k = 0, 1, 2, ..., n-1$

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14. Find the fourth roots of 81.

$$z = 81 + 0i \qquad r = \sqrt{81^2 + 0^2} = 81 \qquad \tan\theta = \frac{0}{81} = 0$$

$$\theta = 0$$

$$w_k = \sqrt[4]{81}\left(\cos\frac{0 + 2\pi k}{4} + i\sin\frac{0 + 2\pi k}{4}\right),\ k = 0, 1, 2, 3$$

$$k = 0: w_0 = 3(\cos 0 + i\sin 0) = 3(1 + i(0)) = 3$$

$$k = 1: w_1 = 3\left(\cos\frac{2\pi}{4} + i\sin\frac{2\pi}{4}\right) = 3\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right) = 3(0 + i(1)) = 3i$$

$$k = 2: w_2 = 3\left(\cos\frac{4\pi}{4} + i\sin\frac{4\pi}{4}\right) = 3\left(\cos\pi + i\sin\pi\right) = 3(-1 + i(0)) = -3$$

$$k = 3: w_3 = 3\left(\cos\frac{6\pi}{4} + i\sin\frac{6\pi}{4}\right) = 3\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right) = 3(0 + i(-1)) = -3i$$

Fourth roots: $\pm 3, \pm 3i$

15. Find the cube roots of $8\left(\cos 204° + i\sin 204°\right).$ Write the results in polar form.

$$w_k = \sqrt[3]{8}\left(\cos\frac{204° + 360°\,k}{3} + i\sin\frac{204° + 360°\,k}{3}\right),\ k = 0, 1, 2$$

$$k = 0: w_0 = 2\left(\cos\frac{204°}{3} + i\sin\frac{204°}{3}\right) = 2\left(\cos 68° + i\sin 68°\right)$$

$$k = 1: w_1 = 2\left(\cos\frac{204° + 360°}{3} + i\sin\frac{204° + 360°}{3}\right) = 2\left(\cos 188° + i\sin 188°\right)$$

$$k = 2: w_2 = 2\left(\cos\frac{204° + 720°}{3} + i\sin\frac{204° + 720°}{3}\right) = 2\left(\cos 308° + i\sin 308°\right)$$

} 3 roots

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